Old Fort, NC – Utility map: note on data gaps

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Deliverables: web map (https://anis00.github.io/nc-utility-map/oldfort/), GeoPackage in NC State Plane feet (EPSG:2264), GeoJSON copies (EPSG:4326), and a sources table. Every feature carries `source`, `source_date`, `capture_method` and `confidence`. Nothing was drawn without a source.

Study area: town limits (NCDOT) plus a 1-mile buffer, plus 1-mile corridors along US-70 and I-40 extending about 3 miles beyond town limits.

Water and sewer

- **Mains, hydrants and manholes are not mapped.** A public records request was sent to the Town on October 5 (follow-up October 8). The Town works from paper and PDF plans prepared by several engineering firms, and these had not been collected at delivery. They can be added to the same GeoPackage and web map when available.
- **Mapped:** the 4 Town wells (#2, #5, #9, #10) from the state Public Water Supply Water Sources layer, checked on aerial imagery; their depths match the 2025 Local Water Supply Plan. The wastewater treatment plant (NPDES NC0021229) from EPA ECHO, moved about 150 ft onto the plant visible on imagery.
- **System figures (2025 Local Water Supply Plan):** water supply 0.645 MGD, average use 0.32 MGD, about 13 miles of lines (2 to 12 in), 608 connections, no emergency interconnection. Treatment plant: 0.900 MGD permitted, 0.140 MGD average, 0.800 MGD maximum day. 262 water customers are on septic systems.
- **Storage tanks not located.** The Town reports 0.70 MG of storage (2 tanks in 2003); they were not found in public data or on imagery. Two tanks mapped in OpenStreetMap about 4 miles west, outside the study area, appear to serve a private system and are not included.
- **2003 service areas** (NC Rural Center survey) are shown for reference only. Marion's water service area reached into the eastern corridor at that time.
- **The system is changing:** the Town's post-Helene program includes replacing about 23,600 ft of water lines and 5,500 ft of sewer. Older plans may not reflect current conditions.
- The plant shows "violation identified" in EPA ECHO for 12 of the last 12 quarters (not significant noncompliance).

Electric

- Service territories (HIFLD, source date 2023-08-01) are approximate. They show Duke Energy Carolinas, Duke Energy Progress and French Broad EMC. **Rutherford EMC serves McDowell County but is not in the HIFLD layer here.** Confirm the provider for each address.
- Transmission lines (HIFLD) date mostly from 2014; all voltages are inferred by HIFLD. Substations come from OpenStreetMap and were checked on imagery.

Broadband

- FCC National Broadband Map, June 30, 2025 data, fixed service reported by providers at census block level, not verified per address. Satellite service (available everywhere) is excluded. Some providers did not report speeds. Census blocks with no serviceable locations are labeled as unpopulated.

Base data

- Parcels are county tax parcels (NC OneMap), not survey-grade. Address points are the county NG911 points. Town limits come from NCDOT and are not a legal survey.